smart contract for storing multidimensional image processing or for long-running, complex computations.

In this chapter, let’s put all these concepts together, not only to reinforce the concepts discussed, but also to comprehend how and why blockchain programming is different from traditional web application development.

Blockchain programming is not about porting a program in any high-level language, such as Java, to a smart contract written in Solidity or a similar language. It is about carefully choosing the data and transactions to be recorded and the rules for validation and verification. This chapter will demonstrate the importance of choosing appropriate data structures, control structures for verification, and notification. You’ll investigate a real educational certificate application. You can apply the knowledge and skills learned in earlier chapters to design, develop, test locally, deploy publicly, and interact with a decentralized solution. End-to-end development is summarized using a real-world use case. The treatment in this chapter will provide an A-to-Z path for Ethereum-based decentralized application development guided by a roadmap. You can download the completed codebase that illustrates all the steps and follow along with the instructions.

Let’s begin with the motivation for the scenario I chose for this chapter.

11.1 Motivating scenario: Educational credentialing

Educational credentialing is a vast problem domain of high interest to many all over the planet. It has diverse stakeholders